

Astrometric Resolution

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Among the stars known to host one or more planets, roughly 25% have a binary stellar companion. Most of these binaries have separations of 100-10,000 AU, but a handful have hazardously close separations of 20 AU or less. The presence of a close binary companion can inhibit or preclude giant planet formation. I will discuss the hypothesis that some fraction of planet-hosting stars that are now in binaries were initially single. This idea is motivated by recent evidence that most stars, including the Sun, were born in clusters with hundreds to thousands of constituents. In a cluster environment, a single star, perhaps with a planet in tow, can acquire a binary companion during a three-star dynamical interaction. Similar dynamical processes can dissolve wide binaries. As an example, I will assess the possibility that the Sun had a distant companion, and consider what dynamical imprint that star may have left on the Solar System.

Consider a circularly symmetric point-spread function (PSF), $f(|\mathbf{x} - \mathbf{x}_c|)$, for a source at position $\mathbf{r}_c$ on the sky. Let $r = |\mathbf{r} - \mathbf{r}_c|$, and suppose that $f(r)$ is peaked at $r = 0$ and is characterized by a FWHM of η . Two sources with the same flux are resolvable if the separation between the centroids of their respective PSFs is larger than $\sim \eta$. Thus, η sets the *angular resolution*. Astrometry is the task of measuring relative angular separations between resolved sources. The precision of astrometric measurements depends on how well the positions of the PSF centroids can be determined. In general, this is a two-dimensional problem. Here we consider the equivalent one-dimensional problem in order to illustrate the relationship between astrometric and angular resolution given some finite measurement error. Suppose the one-dimensional PSF, $f(x)$, is symmetric about x_c , and that $f'(x_c) = 0$. The Taylor expansion of $f(x)$ about x_c is given by

$$f(x) = f(x_c) + \frac{1}{2}(x - x_c)^2 f''(x_c) + O(x - x_c)^3 . \quad (1)$$

Near the centroid, the decline of $f(x)$ as x deviates from x_c depends on the curvature of $f(x)$ via the second derivative $f''(x_c)$. The question is, how far must x deviate from x_c before the drop in $f(x)$ is, in some sense, measurable? Suppose that ϵ is the fractional error in the fit to the observed PSF near the fitted centroid position. An estimate of the error, δx_c , in the measured centroid position is obtained directly from eq. (1), where we substitute $|f(x)/f(x_c) - 1| = \epsilon$ and $x - x_c = \delta x_c$. Keeping only up to second order in $(x - x_c)$, we find the simple result

$$\delta x_c \simeq \left(\frac{2\epsilon f(x_c)}{|f''(x_c)|} \right)^{1/2} . \quad (2)$$

As specific example, consider a Gaussian one-dimensional PSF:

$$f(x) = e^{-(x-x_c)/2\sigma^2} , \quad (3)$$

where A is some normalization constant. The parameter σ is related to the FWHM, Δ , by

$$\sigma = \Delta / \sqrt{2 \ln 2} \simeq 0.85 \Delta \quad (4)$$

The second derivative of $f(x)$, evaluated at $x = x_c$, is given by

$$f''(x_c) = -\frac{f(x_c)}{\sigma^2} , \quad (5)$$

and thus the astrometric error, from eq. (2), is

$$\delta x_c \simeq \sigma \sqrt{2\epsilon} . \quad (6)$$

Letting $\epsilon = \epsilon_{-2} 0.01$, we find that for this particular example

$$\delta x_c \simeq \frac{1}{7} \sigma \simeq \frac{1}{8} \Delta . \quad (7)$$

- Relate ϵ to the source luminosity.
- Show a sample plot of a measured PSF with noise.
- Consider the more general two-dimensional case, but still specialize to the circularly symmetric case.